TINA REZVANIAN

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SUMMARY

Machine Learning Applied Scientist with strong background in Machine Learning, Deep Learning, Statistical Learning and Operations Research, Network Optimization and Algorithm Design. Experienced in design and testing experiments, development, implementation, and automation of pipelines in computer vision, human-computer interaction, face recognition, bounding box and mask detection, personalized recommender systems, and users behavior prediction.

EDUCATION

Harvard Extension School — Online
Graduate Degree, Deep Learning — 2019

Northeastern University — Boston, MA
Ph.D., Industrial Engineering and Operations Research | Minor in Mathematics — 2013 - 2019

Northeastern Illinois University — Chicago, IL
MBA, Master of Business Administration — 2011 - 2013

University of Science and Technology of Mazandaran — Babol, Iran
B.Sc., Industrial Systems Engineering — 2006 - 2010

SKILLS

Technical Convolutional Neural Networks, Recurrent Neural Networks, Long Short-Term Memory, Auto-Encoders, Tree-based Methods, Support Vector Machines, Ensemble Learning, Feature Engineering, Regularization, Personalization & Recommender Systems, Clustering, Principal Component Analysis, A/B Testing, Network & Advanced Optimization, Supply Chain & Logistics, Mixed Integer Programming, Matching Markets, Algorithm Design

Programming TensorFlow, Keras, Python, R, Flask, AWS, MATLAB, CPLEX, SQL, Hadoop

Visualization Matplotlib, Plotly, Seaborn, Pyvis, ggplot2, Gephi, Tableau

WORK EXPERIENCE

Adobe — Sep 2019 - Present
Machine Learning Engineer 4, Document Cloud Data Science — *San Francisco, CA*

- LiquidMode PreSets: Improved readability through ranking document presets with different size, line and character spacing, line height, and character stretch from data collected on virtual readability lab
  - Designed an experiment where users reading speed and comprehension are measured as paragraphs' presentation styles (presets) are manipulated
  - Predicted the top three presets across different user segments by developing a fast and sequential neural network base-model with loss function designed to minimize cost to move the mass of predicted distribution to grand-truth distribution
  - Optimized the ordering within the top three presets using a list-wise learning model that leverage position-aware ListMLE
- FontMART: Used font and reader characteristics to train a font recommender that automatically orders a set of eight fonts per participant by predicting reading speed
  - Developed and compared two pair-wise learning to rank models including a two-tower neural network and a LamdaMart and evaluated their performance using distance and ordered-class based cost functions

- Identified optimal fonts for user segments
- Compared users' preferred fonts, best-performing fonts, and predicted ordered fonts and identified the impact of font characteristics on their alignment

- LM gazeNet: Predicted user eye-gaze on documents with different PDF structures to optimize Liquid Mode conversion from data collected on virtual readability lab
  - Designed an experiment where users' eye-gaze are recorded using an in-house java-based eye tracker as paragraphs' presentation styles (presets) are manipulated
  - Developed convolutional neural networks model to predict coordinates of eye-gaze on documents before liquid mode conversion
  - Interpreted the relationship between eye-gaze coordinates and document presets using a tree-based regression model
- Multi-Channel Tool Attribution: Increased net conversion rates by in-app recommendation for the next best actions and best communication times
  - Developed a multi-head architecture using attention blocks, LSTMs, and fully connected layers to determine the contribution credit of each action on multiple desired outcomes
  - Estimated the credit of actions, days, and action timesteps on user returning, conversion, and attrition
- CausalTIM: Simulated a Randomized Control Trial (aka. A/B testing) using observed data to estimate the causal impact of tools on desired outcomes for multiple user cohorts
  - Investigated users' engagement behavior data to capture features that influence conversion to paid and impact or affected by tool usage
  - Developed separate models to estimate the causal effects using propensity score matching
  - Analyzed results to ensure causality and and investigated confounding effects using attribution models
- Mobile RealTimeRec: Prototyped a real-time deep reinforcement learning recommender technology on device
  - Predicted general users' behavior by developing a generative adversarial network (GAN)
  - Trained a general recommender using deep reinforcement learning
  - Simulated interactions between users and the recommender system
  - Personalized the recommender online using on-policy deep reinforcement learning
  - Improved user experience by 87% compared to traditional deep learning models
- TIM : Improved Acrobat user engagement by developing an unsupervised framework for desktop, mobile, and web Tenured-users' Impact Modeling (TIM)
  - Clustered users based using a Unet auto-encoder architecture
  - Mapped user clusters with users' engagement and attrition data
  - Personalized in-app push notifications and email communications by predicting the implicit rating of tools within each cluster using matrix factorization

Harvard Extension School Nov 2019 - May 2020
Deep Learning Student *Online*

- Face Recognition: Application of convolutional neural networks in matching extracted faces in images to a known/unknown identity in database of publicly available CASIA-WebFace dataset
  - Re-purposed FaceNet, DeepFace models and developed a deep and CNN models using feature extraction, transfer learning, fine tuning, data augmentation, regularization, and embeddings
  - Re-purposed Google's one shot learning FaceNet to identify subjects
  - Automated the pipeline to recognize and label all faces in an image

Insight Data Science Jan 2019 - Feb 2019
Data Science Fellow *Boston, MA*

- PickaBasket: Constructing pre-packaged bestseller baskets for online grocery stores
  - Identified patterns in customers' purchase behavior as probabilistic rules on bestseller combinations
  - Improved confidence and reduced randomness using measures of control and interestingness
  - Clustered items into disjoint and overlapping baskets using linkage and Clique finder clustering
  - Built a web application for end-users accessible at pickabasket.com using Python, Flask, and AWS

Wheels UP Private Jet Company May 2018 - Aug 2018
Data Science Intern *New York City, NY*

- Forecasting Daily Flight Demand and Allocating Aircraft to Regions to Minimize Empty Flights
  - Clustered flight movements by designing regional/connectivity based similarity metrics to use in community detection methods such as agglomerative nesting and edge-betweenness algorithms
  - Predicted daily flight reservation by classifying demand patterns using time series cross correlation
- Segmenting Customers for Promotional Pricing of Empty Position Flights
  - Identified critical components of flight reservation patterns using principal component analysis

Northeastern University Dec 2014 - Sep 2019
Research Assistant *Boston, MA*

- Forecasting Emergency Response Demand for Refugees as a Zero-Inflated Skewed Variables at United Nations High Commissioner for Refugees (UNHCR Supported)
  - Determined countries requiring emergency response combining Support Vector Machines and Platt's scaled probabilities from Logistic Regression using R
  - Predicted the amount of ER demand in determined countries using Generalized Linear Models
  - Analyzed the impact of key predictors on refugees emergency response locations and amounts
  - Automated parameter estimation process and improved total running time to less than a minute
- Developing Matching Algorithms Based on Short Preference Lists at World Food Programme (NSF Supported)
  - Introduced the concept of negotiations in multi-periodic markets in a way to preserve the Top Trading Cycle algorithm's priorities
  - Identified value-adding targets to start a negotiations path
  - developed models to guarantee a minimum match quality by adoption of α-stable matches defined over all periods
  - Increased total welfare by 40% and total matched pairs by 80%, and reduced turnover rate by 83%
  - Developed a dynamic decision support tool for the human resources team using Python and Xlwings
- Integrating Emergency and Ongoing Supply Chains at UNHCR in a Multi-Criteria Decision Making Framework (UNHCR Supported)
  - Minimized total costs and time in budget constrained two-stage stochastic and mixed integer models
  - Used Pareto curves to analyze the trade-off between the cost and lead time objectives using CPLEX
  - Reduced total cost by 31%, and improved response time by 18%

- Developed an open source decision support tool for UNHCR supply chains group GLPK and Python

- Improving U.S. Air Force Cadets Retention Rate by Creating a Recommendation System (NSF Supported)
  - Identified desired features of cadets' and billets' in the same duty code using MATLAB
  - Quantified the impact of preference correlations and network structures on match size and quality
  - Evaluated match quality by rank quantile, worst assignment ranking, and utilities of final matchings

Healthcare Systems Engineering Institute (HSYE) Sep 2013 - Dec 2014
Northeastern University, Research Assistant *Boston, MA*

- Scheduling Appointments and Treatment Nurses using Column Generation at Moffit Cancer Center
  - Improved resource utilization by 60% and patient wait time by 35% using CPLEX and MATLAB
- Improving Referral Quality by Optimizing Curbside Threshold at Massachusetts General Hospital (MGH) Neurology department
  - Optimized the logistic regression threshold through bootstrapping and cross validation
  - Quantified model improvements by precision recall trade off using Python

PUBLICATIONS

Readability Research: An Interdisciplinary Approach. Foundations and Trends® in Human–Computer Interaction 16.4 (2022): 214-324.

- Cai, T., Wallace, S., **Rezvanian,T.**, Dobres, J., Kerr, B., Berlow, S., ... Bylinskii, Z. (2022, June). Personalized Font Recommendations: Combining ML and Typographic Guidelines to Optimize Readability. In Designing Interactive Systems Conference (pp. 1-25).
- Beier, Sofie, et al.(2021)
- Jahre,M.,Kembro,J., **Rezvanian,T.**, Ergun.O, Hapnes(2016).*Integrating supply chains for emergencies and ongoing operations in UNHCR*, Journal of Operations Management,45,57-72.
- Jahre, M., Kembro, J., **Rezvanian, T.**, Ergun, O., Haapnes, S.,& Hultberg, M. *Scenario building in UNHCR-Predicting future demand in refugee crises.* In Proceedings of the 5th World Production and Operations Management Conference, POM Havana 2016
- **Rezvanian, T.**, Ergun, O., Jahre, M., Kembro, J., (2020) *Forecasting Emergency Response Demand for Refugees as a Zero-Inflated Skewed variables: Evidence from UNHCR*
- **Rezvanian, T.** Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness. Diss. Northeastern University, 2019.
- **Rezvanian, T.**, Ergun, O., *Two-sided Stable Matching with Negotiations* (submitted to AINA: Advanced Information Networking and Applications)
- **Rezvanian, T.**, Ergun, O., *Deferred Acceptance Algorithms and Aftermarkets for Matching with Short Preference Lists at World Food Programme* (submitted to Journal of Operations Reseach)